

Little Learning, Large Inertia: Deductible Choice in Swiss Health Insurance*

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September 2026

Abstract

Consumers often make suboptimal choices in health insurance markets, yet most of the evidence comes from particular populations. We study an entire national population making the same recurring choice: deductible choice in Switzerland’s mandatory health insurance market, where benefits are standardized by law. Enrollees who first join the market (shorter-tenured) must make an active choice, while those already enrolled (longer-tenured) may default to their prior plan. We match the two groups on canton, gender, age, and health risk, and document a few stylized facts. First, enrollees’ initial choice quality is high but imperfect: 70% of shorter-tenured enrollees hold the cost-minimizing deductible and 11% a dominated one, forgoing CHF 260 a year. Second, over the next four years, optimal rates stay flat and only 15% ever switch. Third, longer-tenured enrollees choose worse than their matched short-tenured peer, by 14 percentage points on both dominated and cost-minimizing choices. Fourth, switching rate falls steeply with age among longer-tenured enrollees but is flat among shorter-tenured ones. Fifth, stakes do not overcome the default: switching rises 7–9 points per CHF 1,000 of forgone savings, then plateaus below 10%, and the same holds when the

*We thank Justin Sydnor and Cameron Ellis for helpful comments and discussions. We also thank participants at the 2026 Annual Meeting of the American Risk and Insurance Association and at the 2022 CEAR-MRIC Behavioral Insurance Workshop in Munich.

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gain is measured in certainty equivalents, so risk aversion cannot explain the inaction. Sixth, suboptimal choice costs CHF 2.3 billion a year, of which CHF 0.86–1.1 billion is cross-subsidy whose removal would require base premiums to rise 3–4%. Our findings suggest that improving choice quality depends less on information or incentives than on the default itself.

Keywords: consumer inertia, deductible choice, health insurance, default effects, market tenure, choice architecture, subscription markets

JEL classification: D12, D91, G22, I13

1 Introduction

The modern economy increasingly runs on subscription and auto-renewal. Consumers hold recurring contracts in media, software, telecommunications, financial services, and insurance, and those contracts renew unless the consumer acts (McCarthy and Fader, 2017). A large literature documents that consumers rarely change their contracts, and attributes this inertia to inattention, switching costs, search frictions, and the pull of defaults (Handel and Schwartzstein, 2018; Brot-Goldberg et al., 2023; Heiss et al., 2021). Much of what we know, however, comes from particular populations. The canonical evidence is from Medicare Part D, where every enrollee is over 65 (Ketcham et al., 2012, 2015), from Medicaid (Marton et al., 2017), or from the employees of a single firm (Handel, 2013; Bhargava et al., 2017). Whether these patterns describe consumers in general, or the elderly, the poor, and the employed in particular, remains open.

Switzerland’s mandatory health insurance market is a natural place to answer this question, because it lets us observe an entire population making the same choice. Basic insurance is mandated for every resident. The market therefore spans the entire adult population in a single pool, rather than being segmented by age, gender, or employer. Benefits are fixed by federal law and are identical across insurers and across years; the one financial dimension the enrollee chooses is the annual deductible. The menu has six levels, the premium rebate for a higher deductible is a regulated linear discount capped at 70 centimes per franc of additional exposure, and enrollees may move between levels every year at no cost and without medical underwriting.¹ This lets us ask a question that is hard to pose in most markets: when a recurring financial decision is made about as simple as regulation can make it, do consumers choose well, and does the quality of their choices improve with experience or erode as a default takes hold? The answer matters beyond Switzerland, because simplifying menus and relying on repeated choice are among the remedies most often proposed for poor decisions in insurance markets (Abaluck and Gruber, 2023; Handel and Kolstad, 2015).

We use administrative records covering the entire enrolled population from 2017 through 2022, which report each enrollee’s deductible, premium, and medical spending. We compute what each of the six deductibles would have cost every enrollee in every year and identify the cost-minimizing one. The data also contain a natural comparison: Everyone entering the market in 2018 (shorter-tenured) was required to submit an active deductible election

¹The Dutch basic insurance market shares the compulsory, community-rated structure and has been used to study inertia and selection (Croes et al., 2025). It differs in how the deductible is priced: Dutch enrollees choose a voluntary deductible on top of a government-set mandatory amount, and the premium discount attached to each level is set by each insurer rather than by a regulated schedule.

on entry, while enrollees already in the market (longer-tenured) are renewed into last year’s deductible unless they file a change.²

A natural concern with this comparison is that the 2018 entrants are largely recent immigrants, who may differ from the mostly Swiss-born longer-tenured population in ways that matter for choice quality. We address this in two steps. First, we match each shorter-tenured enrollee to a longer-tenured enrollee in the same canton, gender, and age group with the closest health risk, so that every comparison in the paper holds these observable characteristics fixed. Second, in a subsample linked to the Swiss Survey on Income and Living Conditions, foreign and Swiss nationals who report the same subjective health status have similar medical spending, which suggests that nationality does not proxy for unmeasured health within matched cells. Matching cannot address selection into migration itself; we discuss this in Section 3 and read the level gap between the two groups as an upper bound on inertia.³ Conditional on these controls, the two groups differ in two ways that work against each other. Longer-tenured enrollees hold a deductible chosen years earlier and might renew automatically since, whereas the 2018 entrants have just chosen theirs; but longer-tenured enrollees have also had many more years to learn how the market works (Ketcham et al., 2012). Differences in choice quality between the two groups therefore reflect a combination of inertia and learning.

We begin with the 2018 cohort in its first full year in the market. We find relatively high initial choice quality: about 70% choose the optimal deductible and only 11% select a dominated option. Nonetheless, the mistakes are economically significant. Average foregone savings amount to roughly CHF 260 per enrollee per year, and the upper tail is substantial, with a non-negligible share of individuals incurring losses exceeding CHF 1,500. Tracking this cohort over time, we find little evidence of short-run learning. Optimal choice rates remain flat, dominated choices decline only slightly, and average foregone savings do not decrease over the four years following entry. This lack of improvement coincides with low switching rates—around 5–7% per year—suggesting that inactive choice behavior sets in quickly after market entry.

At the same time, focusing on shorter-tenured enrollees who do switch reveals a more nuanced pattern. Among shorter-tenured enrollees who change deductibles, switching is

²We do not observe enrollment histories prior to 2017 and therefore cannot measure tenure precisely among longer-tenured enrollees; because enrollment is mandatory, they may have been in the market for decades.

³If migrants are positively self-selected on attributes correlated with decision quality (Borjas, 1987), then shorter and longer-tenured enrollees’ choice quality difference may reflect both the positive selection and inertia.

typically beneficial: 63–65% experience positive savings, with average gains of CHF 100–250 per switch. Over the period from 2019 to 2022, about 15% of shorter-tenured enrollees ever switch plans, achieving average annual savings of CHF 117. These gains, however, are modest relative to those realized by longer-tenured enrollees, who switch less frequently but benefit more when they do. This contrast suggests that while short-run learning is limited at the population level, the ability to profit from active re-optimization may improve only over a longer horizon.

We then study deductible choices among longer-tenured enrollees, who face the same choice menu but may passively remain in their previously selected plans. Longer-tenured enrollees exhibit substantially worse choice quality than shorter-tenured enrollees. Even after matching, longer-tenured enrollees are 14 percentage points more likely to choose dominated deductibles and 14 percentage points less likely to choose the lowest-cost option. These differences persist over time and imply sizable financial consequences. Foregone savings are significantly higher among longer-tenured enrollees, particularly in the upper tail of the distribution, where losses exceed those of shorter-tenured enrollees by more than CHF 250 per year. Because greater experience would be expected to improve decision-making, these patterns provide suggestive evidence that default behavior and inertia play an important role in shaping insurance choices.

We further document that inertia is unevenly distributed across the population. Switching rates decline monotonically with age among longer-tenured enrollees, reaching their lowest levels among individuals aged 65 and above, even though the potential gains from switching do not decline monotonically with age. In contrast, switching behavior among shorter-tenured enrollees is remarkably stable across age groups, persisting up to age 80. The comparison suggests that the decline in switching with age is more attributable to cumulative market tenure than to declining cognitive ability, the usual reading of age gradients in financial decision quality ([Agarwal et al., 2009](#)). This finding refines the policy challenge: rather than focusing exclusively on elderly enrollees, effective interventions should address all enrollees early in their market tenure, before prolonged default exposure entrenches inaction.

We then examine the underlying mechanism of the inertia we document. The literature broadly distinguishes two classes of explanations for poorly informed choices: frictions and mental gaps ([Handel and Schwartzstein, 2018](#); [Brot-Goldberg et al., 2023](#)). We show that switching behavior responds positively to financial incentives: a CHF 1,000 increase in potential savings raises switching probabilities by 7–9 percentage points. However, the response is highly non-linear: switching rises with the incentive only up to the median and

then settles below 10% even when potential gains are large. The same pattern holds when the incentive is measured in certainty equivalents rather than in realized cost, so risk aversion cannot account for the inaction. These findings indicate that while consumers respond to incentives at the margin, substantial inaction persists even when the stakes are high.

Finally, we quantify the aggregate cost of inertia through counterfactual exercises. We estimate that the Swiss market leaves approximately CHF 2.3 billion per year on the table due to suboptimal deductible choice—roughly 6–7% of the total cost enrollees bear. However, as [Handel \(2013\)](#) demonstrates theoretically, consumer inertia can stabilize risk pools by preventing adverse selection. We confirm this mechanism empirically: because over-insured enrollees currently cross-subsidize the risk pool through excessive premiums, eliminating inertia would drain approximately CHF 860 million–1.1 billion in implicit subsidies and require a base premium increase of roughly 3–4% to maintain market equilibrium.

Our findings have several implications for insurance market design and the broader economics of subscription markets. First, the absence of learning in the short term among shorter-tenured enrollees suggests that policies relying on repeated choice opportunities or passive experience are unlikely to substantially improve consumer outcomes. Second, the large and persistent foregone savings observed among longer-tenured enrollees highlight the importance of default rules. Because defaults exposure accumulates with tenure, enrollees most likely to hold a deductible chosen years earlier also have a longer time for health risk to change, making the default hurting them most ([Bundorf et al., 2012](#)). Third, while switching responds to financial incentives, the response is weak and highly non-linear, limiting the effectiveness of price signals or information disclosure alone. Instead, our results point to the value of interventions that directly address inertia, such as structured re-enrollment, default re-optimization, or mandatory active choice intervals. While our evidence is specific to the Swiss health insurance market, the underlying mechanisms are likely relevant to other settings that rely on passive auto-renewal, such as telecommunications or financial products, though the magnitude of inertia may differ depending on product complexity, switching costs, and supply-side frictions.

Our paper contributes to three strands of literature. First, we add to the literature examining consumer decision-making under conditions of active choice. A growing body of evidence documents the persistent selection of dominated plans across diverse insurance environments ([Handel, 2013](#); [Bhargava et al., 2017](#); [Liu and Sydnor, 2022](#)) and links these errors to bounded rationality and information frictions ([Abaluck and Gruber, 2011](#); [Bhargava et al., 2017](#)). While recent studies often attribute suboptimal selection to the complexity of

the choice environment or “choice overload” from large menus ([Abaluck and Gruber, 2023](#)), we show that poor decision-making persists even in a “lean” choice setting. In our context, benefits are standardized and optimal deductibles are easily identifiable, yet consumers still exhibit significant “choice attenuation.” This implies that the roots of poor decision-making go deeper than mere environmental complexity.

Second, our work contributes to the literature on consumer inertia and the decomposition of friction costs by examining the dynamic evolution of choice quality. While seminal evidence from Medicare Part D ([Handel, 2013](#); [Ketcham et al., 2015](#); [Heiss et al., 2021](#)) and Medicaid ([Marton et al., 2017](#)) focuses on specific elderly or low-income populations, we study a demographically diverse population across the entire age spectrum. This allows us to show that the age gradient in inertia is driven by tenure rather than age per se, and that inaction sets in rapidly after market entry across all age groups. Crucially, our setting allows us to isolate the role of inattention from other structural barriers. Unlike the Medicare environment where switching can be hindered by supply-side frictions—such as changes to drug formularies or provider networks ([Ketcham et al., 2015](#))—the Swiss market features standardized benefits that are independent of the provider. In a setting where benefits are standardized and do not vary across insurers or across years, we find that choice quality does not improve with experience, contrasting with the “learning-to-swim” effects documented by [Ketcham et al. \(2012\)](#). Furthermore, our finding that switching behavior responds positively—though insufficiently—to financial incentives provides a nuanced contrast to [Brot-Goldberg et al. \(2023\)](#), suggesting that while inertia is powerful, it is not entirely attributable to mental gaps: consumers retain some sensitivity to financial stakes, yet this sensitivity plateaus well before inertia is overcome.

Third, we add to a body of work on deductible choice in the Swiss market, which has examined how premium subsidies and information provision shape plan choice ([Kaufmann et al., 2017, 2018](#); [Vaidya, 2021](#); [Boes and Liu, 2024](#); [Biener and Zou, 2024](#)). We complement this work by following the entire enrolled population over five consecutive years, which lets us study not just what determines the deductible but how rarely it changes: fewer than seven percent of enrollees switch in a given year.

The remainder of the paper is organized as follows. Section 2 describes the institutional background. Section 3 presents the data and empirical methodology. Section 4 reports the empirical results. Section 5 presents counterfactual analysis and policy implications.

2 Institutional Background

Switzerland operates a system of mandatory health insurance. All residents are required to purchase compulsory basic health insurance, which provides a standardized package of covered services defined by federal law. While benefits are identical across plans, insurance contracts differ along three main dimensions: insurance company, deductible level, and insurance model (i.e., network type).

Adults aged 19 and above can choose from six vertically differentiated deductible levels: CHF 300, CHF 500, CHF 1,000, CHF 1,500, CHF 2,000, and CHF 2,500, as summarized in [Table 1](#).⁴ Once the deductible is exhausted, enrollees face a coinsurance rate of 10 percent on additional medical expenditures, subject to an annual cap of CHF 700.

Premium differences across deductible levels are regulated. Specifically, the premium rebate for higher-deductible plans relative to the default plan (CHF 300 deductible) is capped at 70 percent of the incremental deductible. For example, choosing a CHF 1,000 deductible instead of the CHF 300 deductible implies a maximum allowable premium reduction of

$$0.7 \times (\text{CHF } 1,000 - \text{CHF } 300) = \text{CHF } 490.$$

In practice, insurers almost uniformly apply a linear rebate schedule equal to this regulatory maximum, such that the premium reduction is proportional to the increase in the deductible. This rebate structure is highly homogeneous across insurance providers. Each year, insurers submit premiums for the default deductible plan (CHF 300) to cantonal regulators for approval. Once approved, premiums for all higher deductible options are mechanically determined according to the linear rebate schedule shown in [Table 1](#).

Premiums are community-rated within age groups. Among adults, there are two age categories: young adults aged 19 to 25, and adults aged 26 and above. Within each age group, premiums do not vary with individual health risk. Young adults face lower community-rated premiums than adults aged 26 and above, but both groups face the same choice menu over deductibles and insurance models as presented in [Table 1](#).

In addition to deductible choice, consumers can select among four main insurance models: the standard (basic) model, Health Maintenance Organization (HMO) models, family doctor models, and TelMed models. Importantly, the scope of covered benefits under compulsory

⁴For children (i.e., individuals aged 18 or below), seven deductible levels are available: CHF 0, CHF 100, CHF 200, CHF 300, CHF 400, CHF 500, and CHF 600.

Table 1: Deductible choice menu

Deductible	Co-insurance rate	Cap co-insurance	Premium (CHF, illustrative)
CHF 300	10%	CHF 700	4,960
CHF 500	10%	CHF 700	$4,960 - (500 - 300) \times 70\% = 4,829$
CHF 1,000	10%	CHF 700	$4,960 - (1,000 - 300) \times 70\% = 4,479$
CHF 1,500	10%	CHF 700	$4,960 - (1,500 - 300) \times 70\% = 4,129$
CHF 2,000	10%	CHF 700	$4,960 - (2,000 - 300) \times 70\% = 3,779$
CHF 2,500	10%	CHF 700	$4,960 - (2,500 - 300) \times 70\% = 3,429$

Notes: The CHF 4,960 annual premium shown for the CHF 300 deductible is illustrative. Insurers set the premium for the default deductible and cantonal regulators approve it, so its level varies by insurer, canton, age group and insurance model; premiums at every higher deductible are then determined mechanically by the regulated rebate shown in the final column.

health insurance is identical across all models. Under the standard model, consumers have unrestricted access to physicians and hospitals. In HMO models, enrollees are required to seek care within a designated physician network or group practice and must consult this network as the first point of contact when illness occurs; in return, they receive a premium discount. Under the family doctor model, enrollees must first consult a designated primary care physician—chosen from the insurer’s approved list—before accessing specialist care. This model typically offers a premium discount of approximately 15 to 20 percent relative to the standard model. Finally, under the TelMed model, enrollees are required to contact a 24-hour medical call center prior to visiting a physician, in exchange for lower premiums. The four models are vertically differentiated by how far they restrict the choice of provider, and premiums are ordered accordingly: the standard model imposes no restriction and carries the highest premium, while the family doctor, TelMed and HMO models each route the first contact through a designated gatekeeper and each carry a discount relative to it.

A basic insurance contract is therefore a triple: an insurer, an insurance model, and a deductible. An enrollee may change any of the three at the annual renewal. Benefits are standardized across all three dimensions, but premiums for that identical coverage differ substantially across insurers and across models, as well as across deductibles.

3 Methodology and Data

3.1 Data

We use individual-level administrative data from the Swiss Federal Office of Public Health, covering the entire population residing in Switzerland between 2017 and 2022. The data

include basic socio-demographic characteristics (age and gender), canton of residence, the enrollee’s insurer, insurance model, and deductible, annual medical expenditures, and insurance premiums. A key limitation of the data is that we do not observe enrollees’ full enrollment histories prior to 2017. As a result, we can identify new market entrants only if an enrollee ID first appears in the data after 2017.⁵

We apply several sample restrictions to construct our baseline analysis sample. First, we restrict attention to adults aged 26 and above who face a common choice menu and are ineligible for premium discounts or subsidies.⁶ Second, we classify individuals into two groups based on enrollment timing: (i) longer-tenured enrollees, defined as individuals who joined the market before 2017, and (ii) shorter-tenured enrollees, defined as individuals whose enrollee ID first appears in 2018.

We further require full-year enrollment. Longer-tenured enrollees must be continuously enrolled from 2018 through 2022; shorter-tenured enrollees must have no enrollment record in 2017, enter in 2018, and be continuously enrolled from 2019 through 2022. We impose this restriction because our choice-quality measures are annual: foregone savings compare the annual cost of the chosen deductible with that of the cost-minimizing one, and both the premium and the cost-sharing schedule are defined over a full calendar year. For an enrollee observed for part of a year, realized spending is mechanically truncated and the deductible that would have minimized cost cannot be determined. Partial-year enrollees would therefore enter the sample with measurement error correlated with the very quantity we study. The restriction also holds the number of choice occasions fixed across individuals, so that cumulative switching rates and the survival analysis in Appendix A are computed over a common exposure window.

The cost is that we exclude enrollees who enter or leave the market mid-panel, and our results apply to the continuously enrolled population. We cannot sign the resulting selection: dropping mid-panel movers removes people who by construction made an active enrollment decision, which pushes measured inertia up, but it also removes the least settled enrollees, whose plan choice is the least informative about attention to the renewal decision. We therefore report the restriction as a scope condition rather than as a conservative choice.

Our window spans the COVID-19 pandemic, which depressed utilization in 2020 and could

⁵Because enrollment in compulsory health insurance is mandatory in Switzerland, individuals already enrolled before 2017 may have been participating in the market for many years or decades. Shorter-tenured enrollees identified after 2017 are primarily recent immigrants.

⁶Individuals aged 0 to 18 face a different deductible menu, while individuals aged 19 to 25 face the same menu as adults but receive lower, age-specific premiums.

in principle have changed both the incentive to switch and the attention enrollees paid to it. Switching among the 2018 cohort is 5.4% in 2019 and 5.4% again in 2022, and the hazard of a first switch is flat from 2020 onward (Appendix A), so the inaction we document is not an artifact of the pandemic.

3.2 Constructing Comparable Shorter- and Longer-Tenured Samples

Shorter- and longer-tenured enrollees differ along two dimensions. First, they differ in the vintage of the plan they hold. Regulation requires an individual entering the market for the first time to submit an active deductible election, so the 2018 cohort holds a plan it chose deliberately and recently. A longer-tenured enrollee may hold a plan chosen many years earlier, under health circumstances and prices that no longer apply, and renewed automatically ever since. Both groups are auto-renewed in every year after entry; what separates them is how many renewals the held plan has passed through. Second, longer-tenured enrollees have had more time to learn how plan options and cost-sharing rules work. Differences in deductible choices between the two groups therefore reflect inertia and learning working in opposite directions.

Two things follow. First, only one of the two groups has a known tenure. The 2018 cohort’s clock starts when we observe it enter. For the longer-tenured group we know only that enrollment began before 2017, which our data cannot date; the group pools enrollees two years into the market with enrollees who have been in it for decades. “Longer-tenured” is therefore a lower bound, and the group is heterogeneous in precisely the variable we argue is doing the work. Second, what the comparison isolates is the vintage of the held plan—how long ago, and how deliberately, it was chosen—holding the menu, the pricing rule, and the standardized benefits fixed by law. Both groups face an automatic renewal in every year after entry. What differs is that the 2018 cohort’s plan is one year old and was actively elected on entry, while the longer-tenured group’s has been carried forward through many renewals. The gap in choice quality between them therefore measures how much choice quality erodes as a plan ages in place. Learning works against finding such a gap: longer-tenured enrollees know the market better, so if experience dominated they should choose better. They choose 14 percentage points worse.

Beyond these institutional differences, shorter- and longer-tenured enrollees may also differ systematically in demographics and health status. To isolate the role of inertia and learning, we construct a matched sample in which shorter- and longer-tenured enrollees are

otherwise comparable. Specifically, for each shorter-tenured enrollee in the baseline sample, we match one longer-tenured enrollee with the closest observed characteristics. We require exact matches on canton of residence, gender, and age group. Among candidates satisfying these criteria, we select the nearest neighbor based on measures of healthcare risk, including risk scores, indicators for chronic disease diagnoses, indicators for inpatient stays in the previous year, and non-discretionary healthcare spending in 2019. Matching on healthcare needs is particularly important in our context because some choice quality measures—such as foregone savings—are sensitive to underlying health risk.⁷

A potential concern is that shorter-tenured enrollees entering the market after 2017 are predominantly migrants, who may differ systematically from the native Swiss population in education, income, or cognitive abilities. In particular, if migrants are positively self-selected on attributes correlated with decision quality (Borjas, 1987), this could partially account for the higher choice quality we observe among shorter-tenured enrollees. Because migration is a selective process at all ages, such positive selection could also explain the flat age gradient in switching behavior among shorter-tenured enrollees—if migrants who relocate at age 65 are comparably capable to those who relocate at age 30, the absence of an age gradient in this group would reflect selection rather than the irrelevance of age.

Several features of our setting partially mitigate this concern. First, Switzerland’s migrant population is predominantly composed of EU/EFTA nationals from neighboring countries (Italy, Germany, France), who face limited cultural and linguistic barriers (Swiss Federal Statistical Office, 2024). Data from the Swiss Earnings Structure Survey show no large systematic economic disparity between EU migrants and Swiss nationals (Swiss Federal Statistical Office, 2025). Second, our matching procedure strictly controls for canton of residence, age, gender, and healthcare risk, which absorbs much of the observable variation between the two groups. Third, in a subsample linked to the Swiss Survey on Income and Living Conditions, foreign and Swiss nationals who report the same level of health have similar distributions of medical spending, which suggests that nationality does not proxy for unmeasured health within our matched cells (Appendix C).

Nonetheless, we cannot fully rule out that unobservable differences between shorter- and longer-tenured enrollees contribute to the patterns we document. The level difference in choice quality between the two groups should therefore be interpreted as an upper bound on the effect of inertia. The differential age gradient—flat among shorter-tenured enrollees,

⁷We use 2019 as the baseline year for health risk measures because many shorter-tenured enrollees enter the market partway through 2018, making risk scores and utilization measures in that year potentially noisy or incomplete.

steeply declining among longer-tenured enrollees—is suggestive of a tenure mechanism, but we acknowledge that migrant selection could also contribute to this pattern.

A related limitation is that we do not observe the initial deductible choices of longer-tenured enrollees. As a result, we cannot distinguish between two possible sources of their lower choice quality: (i) initially appropriate deductible choices that became suboptimal as health needs evolved over time, and (ii) initially poor choices that were never corrected. Both channels are consistent with the inertia mechanism we document—in case (i), inertia prevents re-optimization in response to changing needs, while in case (ii), inertia perpetuates initial mistakes. Our data do allow us to track the 2018 cohort over four subsequent years, and we find little evidence of learning or adjustment, consistent with inertia setting in quickly regardless of initial choice quality.

Table 2 presents summary statistics for both the raw and matched samples. The two groups differ sharply on age and health before matching, which is why every comparison we report below is made on the matched sample. In the unmatched data, shorter-tenured enrollees are, on average, younger and healthier than longer-tenured enrollees. After matching, the two groups are well balanced along observed demographic and health dimensions. Despite this balance, their deductible choices differ: longer-tenured enrollees are more likely than shorter-tenured enrollees to choose low-deductible plans.

3.3 Constructing Choice Quality Measures

Our analysis focuses on a specific dimension of plan choice: deductible selection. As described in the institutional background, all insurers offer the same set of deductible options with identical cost-sharing rules. We study the deductible rather than the insurer or the model because it is the only one of the three dimensions on which we can say whether a choice was a good one. We observe all three. But the value of an insurer or an insurance model to a particular enrollee depends on things our data do not contain: which physicians and hospitals they want access to, where they live relative to a network, how much they mind consulting a gatekeeper before seeing a specialist, what they make of an insurer’s service. An enrollee paying more on those dimensions may be buying something we cannot see, so a cheaper contract is not necessarily a better one, and a more expensive one is not evidence of a mistake. The deductible carries no such attributes. Benefits are standardized across deductible levels, cost-sharing is set federally, and the premium rebate is a regulated function of the deductible alone, so the entire difference between two deductible choices is financial and can be evaluated against realized spending. This is what makes a dominated deductible

Table 2: Summary Statistics

Sample	Sample before match			Sample after match		
	Shorter	Longer	Difference	Shorter	Longer	Difference
Demographic variable						
Age	42.50 (11.59)	56.34 (16.03)	-13.84*** (0.05)	41.99 (10.88)	41.99 (10.88)	0.00 (0.07)
Female (percentage)	0.48 (0.50)	0.51 (0.50)	-0.03*** (0.00)	0.48 (0.50)	0.48 (0.50)	-0.00 (0.00)
Inpatient hospital state(percentage)	0.02 (0.12)	0.06 (0.25)	-0.05*** (0.00)	0.01 (0.12)	0.02 (0.13)	-0.00*** (0.00)
Chronic disease (percentage)	0.01 (0.07)	0.03 (0.17)	-0.02*** (0.00)	0.01 (0.07)	0.01 (0.08)	-0.00*** (0.00)
non-discretionary medical spending	2011.30 (6270.20)	4076.65 (9167.63)	-2065.35*** (29.49)	1946.66 (6121.72)	1968.11 (5995.40)	-21.45 (40.38)
Healthcare spending	2059.08 (6306.63)	4173.03 (9201.01)	-2113.95*** (29.66)	1993.40 (6157.93)	2020.63 (6031.78)	-27.23 (40.62)
Deductible Choice in 2019						
Deductible 300	0.23 (0.42)	0.44 (0.50)	-0.21*** (0.00)	0.22 (0.42)	0.27 (0.44)	-0.05*** (0.00)
Deductible 500	0.04 (0.20)	0.13 (0.34)	-0.09*** (0.00)	0.04 (0.20)	0.09 (0.28)	-0.05*** (0.00)
Deductible 1000	0.02 (0.16)	0.04 (0.19)	-0.01*** (0.00)	0.02 (0.15)	0.04 (0.19)	-0.01*** (0.00)
Deductible 1500	0.04 (0.19)	0.08 (0.28)	-0.05*** (0.00)	0.04 (0.19)	0.09 (0.29)	-0.05*** (0.00)
Deductible 2000	0.01 (0.11)	0.03 (0.16)	-0.01*** (0.00)	0.01 (0.11)	0.03 (0.18)	-0.02*** (0.00)
Deductible 2500	0.66 (0.47)	0.28 (0.45)	0.37*** (0.00)	0.66 (0.47)	0.48 (0.50)	0.18*** (0.00)
Choice quality						
Dominated plan (rate)	0.12 (0.32)	0.28 (0.45)	-0.16*** (0.00)	0.11 (0.32)	0.25 (0.43)	-0.14*** (0.00)
Optimal choice (rate)	0.69 (0.46)	0.51 (0.50)	0.18*** (0.00)	0.70 (0.46)	0.56 (0.50)	0.14*** (0.00)
Foregone savings (CHF)	258.43 (470.87)	367.22 (507.68)	-108.79*** (2.20)	259.54 (472.45)	385.77 (534.28)	-126.23*** (3.36)
Switch deductible (rate)	0.05 (0.23)	0.05 (0.21)	0.01*** (0.00)	0.05 (0.23)	0.06 (0.24)	-0.01*** (0.00)
Choice stake (CHF)	229.72 (326.99)	412.64 (358.50)	-182.92*** (1.53)	225.86 (324.97)	235.75 (328.88)	-9.89*** (2.18)
Observations	46,236	4,296,440		45,033	45,033	

Notes: This table reports summary statistics for shorter- and longer-tenured enrollees before and after matching. Shorter-tenured enrollees are individuals whose enrollee ID first appears in 2018. Longer-tenured enrollees entered the market prior to 2017. Matching is conducted on age group, gender, canton, and health risk measures. Standard deviations are reported in parentheses.

identifiable as a mistake.

Three limits follow. First, the foregone savings we compute are those available from re-optimizing the deductible, not the total savings available in this market; an enrollee may also be paying more than necessary for an insurer or a model, and we take no position on whether that is a mistake. Second, foregone savings are measured in realized cost and therefore do not price the risk protection that a lower deductible provides. We observe neither enrollees' risk preferences nor exogenous variation that would identify them, so rather than

impose a distribution of risk aversion, we follow the convention of reading foregone savings as money left on the table (Ketcham et al., 2012, 2015). Where risk bears directly on our conclusions, we report a risk-adjusted counterpart built from calibrated certainty equivalents (Section 4). Finally, our switching indicator counts changes in the deductible only, so an enrollee who changes insurer or model while keeping the same deductible is recorded here as a non-switcher.

Determining the optimal deductible is straightforward in the Swiss setting. The most relevant parameter is an enrollee’s medical spending. An enrollee whose annual spending falls below roughly CHF 2,000 minimizes total cost at the CHF 2,500 deductible; one whose spending exceeds CHF 2,000 minimizes it at CHF 300. This cutoff is a product of the pricing rule rather than of preferences: the premium rebate is a linear discount in the deductible, so each additional franc of deductible buys a fixed CHF 0.70 of premium relief, while the offsetting exposure is borne only in states where spending reaches the deductible. The same proportional pricing makes the four intermediate deductibles never cost-minimizing, at any level of spending. Risk aversion is normally a second determinant of cost-sharing choice, but here it too is constrained by the contract pricing: rationalizing the observed choices through risk preferences would require implausible levels of risk aversion, as Biener and Zou (2024) show.

Constructing these measures takes three steps: recover what each enrollee would have spent under a counterfactual deductible, compute the total cost to the enrollee at each of the six deductibles, and compare the chosen deductible with the cost-minimizing one. The first step is the only one requiring an assumption, because moving to a different deductible changes the price of care and therefore the quantity consumed.

To account for moral hazard responses, we follow the framework of Einav et al. (2013). Let t denote an enrollee’s realized non-discretionary medical spending. After observing t , enrollees trade off the utility from additional medical consumption against the marginal cost-sharing implied by their plan. Under this framework, the enrollee’s optimal total medical spending under plan j is given by

$$m(t, j) = \max\{0, t + \omega(1 - c'_j)\},$$

where c'_j denotes the marginal coinsurance rate under plan j , and ω captures the magnitude of moral hazard responses. In our baseline analysis, we set $\omega = 182$, corresponding to the mean estimate reported in Einav et al. (2013), and conduct robustness checks for alternative values of ω .

Given each individual’s observed spending $m(t_i, \tilde{j})$ under the chosen plan $\tilde{j}$, we invert the above relationship to recover t_i . We then use this inferred t_i to compute counterfactual spending $m(t_i, j)$ under all alternative deductible options. For each plan j , cost-sharing is given by $c_j(m(t_i, j))$, where $c_j(\cdot)$ denotes the plan’s cost-sharing schedule. We define the *total cost to the enrollee* under plan j as the sum of this cost-sharing and the plan premium, and we use that term throughout; it is distinct from *gross medical spending*, the value of care consumed regardless of who pays for it. Conditional on t_i , the optimal deductible minimizes the total cost to the enrollee.

Using this framework, we construct three measures of choice quality. First, following [Biener and Zou \(2024\)](#), we classify all deductible choices other than CHF 300 and CHF 2,500 as dominated, and construct an indicator for whether the enrollee chooses a dominated deductible. Second, we construct an indicator for whether the enrollee chooses the optimal deductible. Because enrollees may not perfectly anticipate their future healthcare needs at the time of choice, we consider two benchmarks: a myopic benchmark, in which the optimal deductible is determined by healthcare spending in the previous period, and a perfect-foresight benchmark, in which the optimal deductible is determined by contemporaneous spending. Third, we compute foregone savings, defined as the difference between the total cost to the enrollee under the chosen deductible and that under the optimal deductible.⁸

4 Empirical Analysis

This section presents empirical evidence on how enrollees choose deductibles in the Swiss health insurance market. Guided by the institutional features and choice quality measures introduced in Section 3, we study two groups of individuals who face the same menu of deductible options but differ in their decision environment: shorter-tenured enrollees, who must make an active choice upon entry, and longer-tenured enrollees, who may passively remain in their default plan. This comparison allows us to separately assess the roles of

⁸The dominance classification is robust to the preference model used to evaluate the deductibles. [Biener and Zou \(2024\)](#) examine whether the choice of an intermediate deductible can be rationalized under a range of alternative decision models: expected utility with risk aversion, a liquidity-constrained model in which enrollees cannot smooth out-of-pocket payments within the year, and reference-dependent preferences in which enrollees evaluate outcomes relative to a reference point. None of these changes the dominated nature of the intermediate plans. Under every specification the cost-minimizing choice remains either CHF 300 or CHF 2,500, and the four intermediate deductibles remain dominated for any level of expected spending. Medical spending is therefore the key parameter determining choice optimality: a healthier enrollee whose annual spending is below roughly CHF 2,000 minimizes total cost at the CHF 2,500 deductible, while an enrollee whose spending exceeds this threshold minimizes it at the CHF 300 deductible. This is mainly because the pricing of the deductible menu does not reflect enrollees’ risk: the premium rebate is a fixed proportion of the deductible regardless of an enrollee’s expected spending.

limited attention, learning, and inertia in driving sub-optimal insurance choices.

4.1 Choices of Shorter-Tenured Enrollees and Consumer Learning

We begin by analyzing the deductible choices of the 2018 cohort. Focusing on this group provides a clean benchmark, because these enrollees were required to elect a deductible on entry rather than inheriting one. We evaluate that election using 2019 rather than 2018. Many of the cohort join partway through 2018, so their 2018 election has no well-defined horizon—a deductible chosen in September may be a decision about the remaining months of 2018 or a forward-looking decision about 2019—and their 2018 spending is not comparable to a full year of it. Because the premium and the cost-sharing schedule are both defined over a calendar year, 2019 is the first year in which the election can be scored against a full-year cost comparison. Only 5.4% of the cohort switched deductible between 2018 and 2019 (Table 3), so the 2019 holdings we evaluate are overwhelmingly the entry election itself.

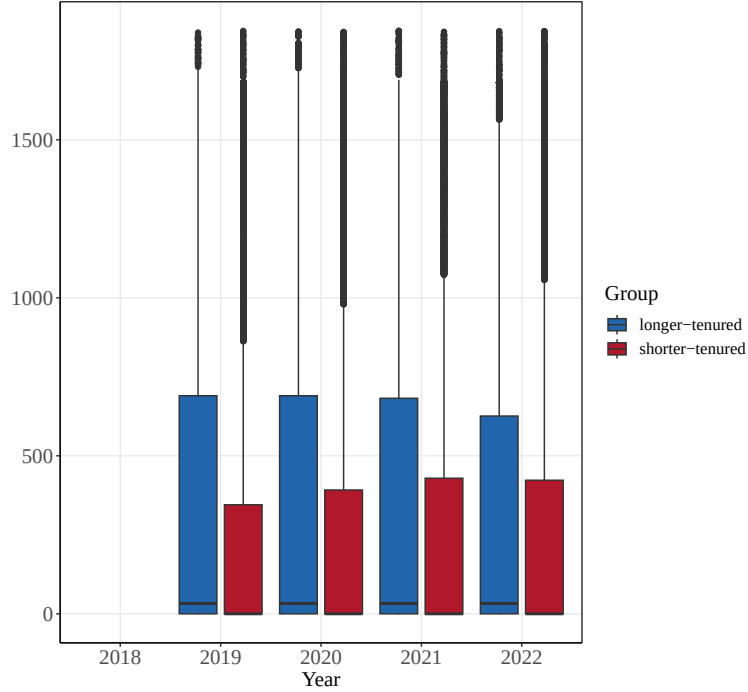
Table 2 shows that shorter-tenured enrollees exhibit relatively high choice quality in their first full year of enrollment. Approximately 70% of shorter-tenured enrollees choose the optimal deductible, while only 11% select a dominated option. Nonetheless, sub-optimal choices are economically meaningful. Average foregone savings amount to CHF 260 per enrollee per year.

Figure 1 (red bars) illustrates the distribution of foregone savings for shorter-tenured enrollees. While the majority of individuals cluster near zero—consistent with optimal choice—the upper tail is substantial. The 75th percentile of foregone savings is around CHF 350, and a non-negligible fraction of enrollees incur losses exceeding CHF 1,500. Table 3 provides the average foregone savings for shorter-tenured enrollees, and the size is around CHF 270, accounting for almost 6% of their total cost. These patterns indicate that even in a setting where the optimal deductible is straightforward to determine, active choice does not guarantee optimality for all consumers.

A key question, then, is whether such suboptimal choices among shorter-tenured enrollees would diminish over time, as would be expected under a learning or mental gap model. If consumers initially make poor choices because they lack knowledge or financial literacy, repeated exposure to the market should lead to systematic improvements in the quality of their choices.

We examine whether learning behavior exists in the short run by assessing how the quality of choice among shorter-tenured enrollees changes over 2019-2022, the four years immediately

Figure 1: Distribution of Foregone Savings



Notes: This figure plots the distribution of foregone savings for shorter-tenured enrollees and longer-tenured enrollees. Foregone savings are defined as the difference between the realized total cost to the enrollee under the chosen deductible and that under the cost-minimizing deductible, conditional on realized health care utilization.

after their initial choice in 2018. Figures 2 and 3 show that over the four years following market entry, the rate of choosing the optimal deductible remains essentially flat, and the share of dominated choices declines only slightly. Consistent with the choice pattern, in Figure 1 (red bars) and Table 3, we find no reduction in foregone savings across the four years. Together, these results indicate little evidence of learning in deductible choice in the short run.

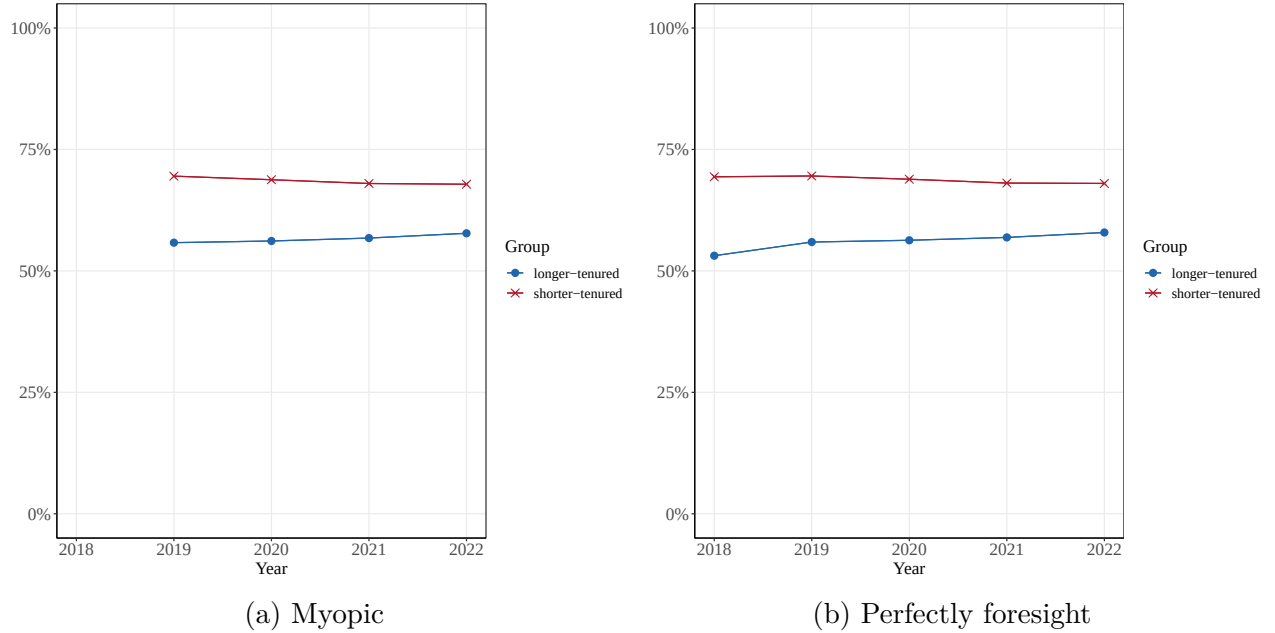
One interesting pattern we highlight is that the lack of learning over time is consistent with the low switching rate among shorter-tenured enrollees. In fact, Table 3 shows that the annual switching rate is around 5.4% to 6.9% among the shorter-tenured enrollees right after they join the market. The numbers are close to the switching rates by longer-tenured enrollees, who have been in the market for many years. It seems that inertia, or inactive choices, sets in quite rapidly when enrollees enter the market. In Appendix A, we further examine the cumulative switching rates. We find that about 15% of shorter-tenured enrollees had ever switched plans four years after entering the market.

Table 3: Spending and Switching for the 2018–2022 Cohort

	2018	2019	2020	2021	2022
Shorter-tenured ($N = 45,033$)					
Switched deductible from the prior year (%)	—	5.4	6.9	6.1	5.4
Total cost to enrollee	4104	4516	4624	4735	4782
Cost at cost-minimizing deductible	3793	4277	4381	4493	4542
Foregone savings	-	260	272	272	272
Foregone savings as percent of total cost (%)	-	5.7	5.9	5.7	5.7
Longer-tenured ($N = 45,033$)					
Switched deductible from the prior year (%)	8.9	6.1	5.7	5.2	5.0
Total cost to enrollee	4586	4673	4774	4864	4897
Cost at cost-minimizing deductible	4219	4319	4436	4540	4590
Foregone savings	-	386	373	359	345
Foregone savings as percent of total cost (%)	-	8.3	7.8	7.4	7.0

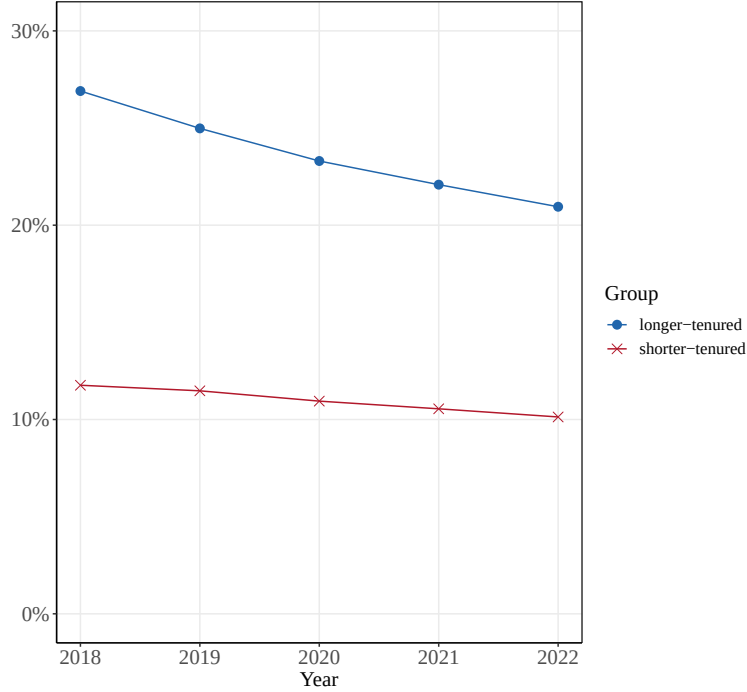
Notes: This table reports the total cost to the enrollee, the minimum attainable cost, and switching behavior for the 2018–2022 cohort. The total cost to the enrollee is the premium plus cost-sharing. Minimum spending is computed using the cost-minimizing deductible under the ex post approach.

Figure 2: Rate of Choosing the Optimal Deductible



Notes: This figure shows the share of individuals choosing the optimal deductible over time. Panel (a) defines optimality under a myopic benchmark, using lagged health expenditures. Panel (b) defines optimality under a perfect-foresight benchmark, using contemporaneous expenditures. Shorter- and longer-tenured enrollees are shown separately.

Figure 3: Rate of Choosing the Dominated (Intermediate) Option



Notes: This figure plots the share of individuals choosing a dominated deductible (intermediate plans) over time.

Despite a low overall switching rate, we also focus on those who switched plans in the four years immediately after the initial choices and examine whether they benefit from switching. If shorter-tenured enrollees gain experience in the market, they should derive positive value from switching. Table 4 shows that, on average, shorter-tenured enrollees benefit from switching plans: among those who switch plans, the mean savings from switching is around CHF 100 to CHF 250, calculated as the difference in the out-of-pocket costs (premium plus cost-sharing) between their chosen plan and the choice last year. We also find that 63%-65% of shorter-tenured enrollees who switch their choices experience positive outcomes from plan switching. In Table 5, we further calculate the cumulative switching and savings from switching from 2019 to 2022. We find that 20% of shorter-tenured enrollees ever switch plans and achieve an average annual savings of CHF 117, representing about 2.2% of their total cost.

We evaluate the economic magnitude of these savings by comparing them with longer-tenured enrollees. We find that longer-tenured enrollees gain more from switching. For example, among longer-tenured enrollees who have switched in the past four years, the mean savings from switching are approximately CHF 260-CHF 450; the cumulative savings from

Table 4: Switching and savings from switching, 2019-2022:

	2019	2020	2021	2022
Shorter-tenured ($N = 45,033$)				
Switched deductible from the prior year (%)	5.4	6.9	6.1	5.4
Ex post approach with perfectly inelastic demand for healthcare				
Mean savings from switching	251	177	119	137
Median savings from switching	462	420	357	416.1
Percent that saved > 0 from switching (%)	65.8	65.3	63.8	65.2
Longer-tenured ($N = 45,033$)				
Switched deductible from the prior year (%)	6.1	5.7	5.2	5.0
Ex post approach with perfectly inelastic demand for healthcare				
Mean savings from switching	448	329	270	260
Median savings from switching	463	446	372	381
Percent that saved > 0 from switching (%)	76.0	72.8	70.9	70.8

Notes: This table reports switching rates and savings from switching among matched shorter- and longer-tenured enrollees from 2019 to 2022. Savings are calculated among the individuals who switched in the year.

switching are CHF 193, accounting for 4% of total cost; and more than 70% of longer-tenured enrollees who switch plans experience positive gains. These findings suggest that learning may occur over the long term. With a longer period in the market, longer-tenured enrollees appear better at navigating options and derive greater benefits from switching.

We also benchmark our results with [Ketcham et al. \(2015\)](#), who find evidence of learning among Medicare Part D enrollees. To directly compare our results with theirs, we also restrict our population to those who aged 65 and present the measures in Table 6. Overall, we find that switching rates in our sample are much lower than those observed among Medicare Part D patients in their context (3%-5% in our setting vs 10%-11% in their setting). The difference may be driven by the complexity of the task (our enrollees must choose basic medical insurance plans, whereas Part D enrollees choose only prescription drug plans) or by other institutional differences between the two markets.

4.2 Choices of Longer-Tenured Enrollees and Inertia

We now turn to the deductible choices of longer-tenured enrollees, who face the same choice menu but operate under a different institutional environment. Longer-tenured enrollees have been renewed into a previously selected deductible many times over. As a result, their observed choices may reflect accumulated inertia rather than deliberate optimization.

At first glance, longer-tenured enrollees appear to exhibit inertia, which results in low

Table 5: Cumulative switching and savings from switching, 2019- 2022:

	2019	2020	2021	2022
Shorter-tenured ($N = 45,033$)				
Percent of current enrollees that have ever previously switched deductible plans (%)	5.4	11.6	16.5	20.3
Actual OOP spending by those who ever previously switched	4853	5031	5185	5234
<i>Ex post approach with perfectly inelastic demand for healthcare</i>				
OOP spending if stayed in the original plan and never switched	4920	5141	5299	5351
Annual savings by current enrollees who ever previously switched	67	110	114	117
Savings as a percent of total cost (%)	1.4	2.2	2.2	2.2
Total year enrollees	9,146	9,146	9,146	9,146
Annual savings from cumulative switching (millions)	0.6	1.0	1.0	1.1
Total cumulative savings from switching (millions)	0.6	1.6	2.7	3.7
Longer-tenured ($N = 45,033$)				
Percent of current enrollees that have ever previously switched deductible plans (%)	14.1	18.4	22.0	25.1
Actual OOP spending by those who ever previously switched	4642	4740	4840	4862
<i>Ex post approach with perfectly inelastic demand for healthcare</i>				
OOP spending if stayed in the original plan and never switched	4752	4899	5019	5055
Annual savings by current enrollees who ever previously switched	109	159	179	193
Savings as a percent of total cost (%)	2.4	3.3	3.7	4.0
Total year enrollees	11,323	11,323	11,323	11,323
Annual savings from cumulative switching (millions)	1.2	1.8	2.0	2.2
Total cumulative savings from switching (millions)	1.2	3.0	5.1	7.2

Notes: This table tracks individuals who have ever switched plans over the years. The annual savings amount that enrollees saved by switching from the deductible choice to the current plan.

choice quality. Table 2 shows that only 6% of longer-tenured enrollees switch deductibles in a given year, and only 56% choose the optimal deductible. However, neither pattern alone is sufficient to establish inertia. Low switching rates could reflect stable health needs, and low optimality rates could reflect persistent knowledge gaps rather than default effects.

To isolate the role of inertia, we compare longer-tenured enrollees to shorter-tenured enrollees who are observationally similar in demographics and health status, as described in Section 3. If time in the market improved decision-making, or even left it unchanged,

Table 6: Switching and savings from switching for aged 65, 2019-2022:

	2019	2020	2021	2022
Shorter-tenured				
Switched deductible from the prior year (%)	5.1	4.0	3.9	3.8
Number of enrollees	2,178	2,471	2,795	3,185
Ex post approach with perfectly inelastic demand for healthcare				
Mean savings from switching	389	124	78	161
Median savings from switching	462	217	217	346
Percent that saved > 0 from switching (%)	73.0	65.0	68.8	75.8
Longer-tenured				
Switched deductible from the prior year (%)	2.8	3.5	3.2	3.2
Number of enrollees	2,178	2,493	2,885	3,243
Ex post approach with perfectly inelastic demand for healthcare				
Mean savings from switching	309	117	197	118
Median savings from switching	372	178	252	147
Percent that saved > 0 from switching (%)	78.7	70.5	72.8	66.0

Notes: This table reports switching behavior and savings from switching for individuals aged 65.

longer-tenured enrollees should choose at least as well as the 2018 cohort.

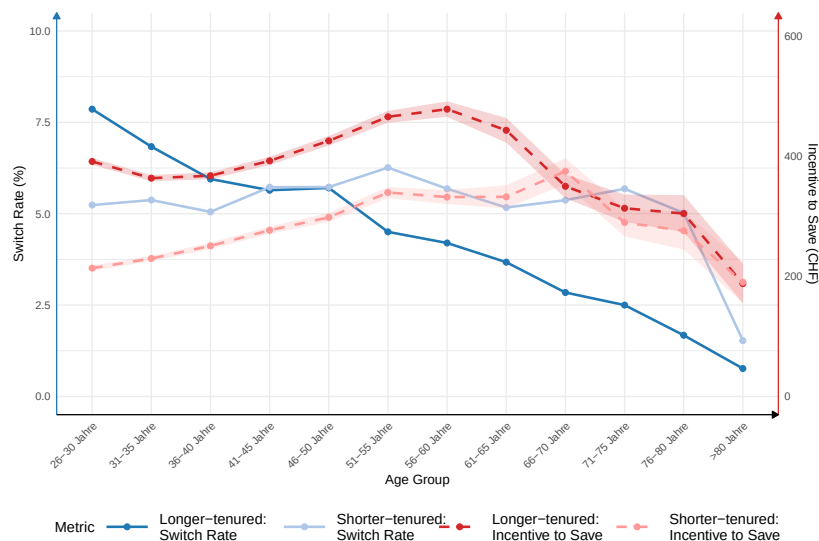
The data strongly reject this hypothesis. Table 2 shows that, even after matching, longer-tenured enrollees in 2019 are 14 percentage points more likely to choose a dominated deductible and 14 percentage points less likely to choose the optimal deductible than shorter-tenured enrollees. Figures 2 and 3 demonstrate that this gap persists over time: longer-tenured enrollees are consistently 15–20 percentage points less likely to choose the optimal deductible under both the myopic and perfect-foresight benchmarks. Because any remaining differences in knowledge or experience would bias longer-tenured enrollees toward better choices, these estimates provide a lower bound on the impact of inertia on choice quality.

The financial consequences of inertia are substantial. Figure 1 compares the distribution of foregone savings between matched shorter- and longer-tenured enrollees. While the median and lower quartile are similar across groups, the upper tail differs sharply. The 75th percentile of foregone savings among longer-tenured enrollees exceeds that of shorter-tenured enrollees by more than CHF 250 per year—an amount comparable to the entire upper-tail loss of shorter-tenured enrollees at entry. For the quarter of enrollees with the worst outcomes, foregone savings reflect roughly equal contributions from initial choice errors (as observed among shorter-tenured enrollees) and inertia-induced failure to adjust. Table 3 reports that the average foregone savings are around CHF 345–CHF 385 for longer-tenured enrollees,

much higher than the values for shorter-tenured enrollees ($\sim$ CHF 270).

Age Dynamics in Switching Behavior We next examine heterogeneity in switching behavior across age groups among longer-tenured enrollees. Figure 4 illustrates a clear monotonic decline in market engagement as enrollees age. While younger cohorts exhibit higher propensities to adjust their coverage frequently, the probability of switching deductibles drops significantly among older enrollees, reaching its lowest point for those aged 65 and above. In contrast, the benefits of switching, measured as the difference in total cost to the enrollee between the default and cost-minimizing options, are not monotonic with respect to age. The key takeaway is that longevity in the market reinforces a ‘sinking’ effect: consumers become increasingly anchored to their initial plans over time. This age-driven inertia is a primary driver of the persistent sub-optimality observed in the longer-tenured population, as older enrollees fail to re-optimize despite changing health risks. Note that the monotonic pattern only exists among longer-tenured enrollees. Among shorter-tenured enrollees, we find a much more stable switch rate across age groups (until age 80). The comparison suggests that the decline in switch rate with age is more attributable to market experience than to cognitive ability.

Figure 4: Switch Rate and Incentive to Save by Age Group (2019)



Notes: This figure illustrates age patterns in switching behavior and incentives to switch. Solid lines plot annual deductible switching rates by age group. Dashed lines plot the incentive to switch, defined as the difference in total cost to the enrollee between the default deductible and the cost-minimizing deductible, conditional on realized utilization. Shorter- and longer-tenured enrollees are shown separately.

4.3 Mechanisms: Frictions versus Mental Gaps

The evidence so far establishes that longer-tenured enrollees choose worse than the 2018 cohort and that the gap does not close with time in the market. It does not say why. This section asks whether the failure to switch is better described as a friction—switching is costly, so enrollees act only when the gain exceeds the cost—or as a mental gap, in which the choice is not reconsidered at all and the size of the gain is therefore irrelevant. The two accounts make different predictions about one object: the slope of switching with respect to the money at stake. Under a friction model that slope is positive and, above the switching cost, steep. Under a mental-gap model it is flat. We construct the money at stake for each enrollee in each year, estimate that slope, and then ask whether the slope is constant across the range of the stakes, which is where the two accounts are hardest to separate and where our data are most informative.

We therefore investigate whether the inertia we document is better explained by friction-based models or mental gap models of default effects (Brot-Goldberg et al., 2023). The key distinction is whether individuals are more likely to switch when the benefits of switching are larger. Under friction models, higher potential gains should increase switching; under mental gap models, switching is largely unresponsive to incentives. We examine this distinction empirically.

4.3.1 Switching Behavior and Incentives to Switch

First, we examine whether enrollees are more likely to switch plans when doing so yields greater benefits. We define the incentive to switch, v_{it} , as the difference in total cost to the enrollee between the default deductible plan and the optimal deductible plan, where that cost is cost-sharing (adjusted for moral hazard) plus the premium. Variation in the incentive to switch across individuals indicates differences in choice quality in the last period or in health needs that change in the current period. Fluctuation of the value within an individual over time indicates changes in health needs.

We first estimate the relationship between switching behavior and this incentive using cross-sectional variation:

$$\mathbf{1}(\text{switch})_i = \beta_0 + \beta_1 v_i + \varepsilon_i, \quad (1)$$

where $\mathbf{1}(\text{switch}_i)$ is a dummy variable indicating whether the individual chooses a different deductible than in the previous year. We estimate equation (1) as a single-year cross-section covering both longer- and shorter-tenured enrollees in 2020—the first year in which both groups enter the baseline panel described below—so that the variation is identified across

rather than within individuals.

The above comparison may not reflect a causal effect of switching behavior on the benefits of switching if the correlation between switching and incentives to switch is driven by other confounding factors, e.g., differences in demographics and other factors between those facing larger and smaller incentives to switch. An ideal experiment would randomly assign individuals to different default plans and compare their switching rates. Lacking such variation, we instead exploit within-individual variation in the incentive over time, using panel data and adding individual fixed effects in the regression, and estimate model (2):

$$\mathbf{1}(switch)_{it} = \beta_0 + \beta_1 v_{it} + \gamma_i + \varepsilon_{it}. \quad (2)$$

Controlling for individual fixed effects ensures that other individual characteristics, including risk preferences, education level, and other unobserved characteristics, do not confound the results. The fixed-effects specification exploits within-person variation in incentives to switch over time by comparing the same individual’s switching behavior under different incentives across years.

Two caveats deserve emphasis. First, the fixed-effects estimate removes time-invariant individual heterogeneity but does not remove serial dependence in the switching decision itself. An enrollee who switched last year has, mechanically, a different default this year, and switching decisions are likely correlated within a person over time for reasons unrelated to the incentive. We cluster standard errors at the individual level throughout, which accommodates arbitrary within-person correlation for inference, but clustering does not address the more basic point that the year-to-year variation in v_{it} comes from realized health shocks rather than from an exogenously assigned default.

Second, and consequently, we do not read β_1 as the causal effect of the default on switching. Identifying inertia in the strict sense requires variation that reassigns defaults independently of the enrollee’s circumstances—a menu change, a plan exit, the creation of a new market, or an experimental default—and our setting supplies none of these. What equation (2) establishes is a weaker but still informative fact: holding the individual fixed, the probability of switching moves with the size of the available gain. This is the comparative static that separates friction models from mental-gap models (Brot-Goldberg et al., 2023), and it is the object the mechanism question in this section requires. We accordingly describe the estimates below as measuring the responsiveness of switching to incentives, not as an estimate of the cost of inertia.

Table 7 shows that switching probabilities increase with the incentive to switch. Panel A measures the incentive as a difference in realized cost, as defined above. A CHF 1,000 increase in potential savings raises the probability of switching by 7–9 percentage points, depending on the specification. Importantly, this relationship persists when we include individual fixed effects, exploiting within-person variation over time and ruling out time-invariant unobserved heterogeneity.

Additionally, we construct a risk-adjusted counterpart of the incentive to switch in three steps. First, we specify the risk distribution as the empirical distribution of realized non-discretionary spending among enrollees in the same year, age class, gender, and previous-year spending bin. Second, we assume constant absolute risk aversion. Third, we set the coefficient of absolute risk aversion to $\gamma = 5.34 \times 10^{-4}$ following Handel (2013), and verify robustness at the higher value implied by Cohen and Einav (2007). The risk-adjusted incentive to switch is then the certainty equivalent of staying in the current plan less the lowest certainty equivalent available.

The main finding still holds. Pricing risk reduces the incentive to switch to roughly 58% of its unadjusted level—part of what a cost-based measure records as foregone savings is genuine insurance value—but switching behavior itself is unchanged. The coefficients in Panel B are larger under the panel specifications (0.12 against 0.09 in columns (2)–(4)) because the same switching is scaled against a smaller measured incentive, not because enrollees act more readily on risk-adjusted stakes.

However, the response is not monotonic. Figure 5 shows the switching rate among those whose default plan is already optimal (incentives to switch being zero) and those with positive incentives to switch. For the latter, we create four equal-sized bins based on the incentives to switch, and we plot the two measures of the incentive side by side. Under the cost-difference measure, switching rates rise with incentives only up to the median incentive level and then fall back to roughly 9%. Under the certainty-equivalent measure the profile is flatter still: switching is essentially constant at about 10% across the first three quartiles and declines to 8% in the highest. Enrollees facing the largest risk-adjusted gains are therefore no more likely to act than those facing modest ones. Risk aversion cannot account for this inaction, since the incentive against which enrollees are failing to respond already prices the reduction in risk exposure that a lower deductible provides. This pattern suggests that, although some inertia is consistent with rational frictions, substantial inaction persists even when switching would yield substantial financial benefits.

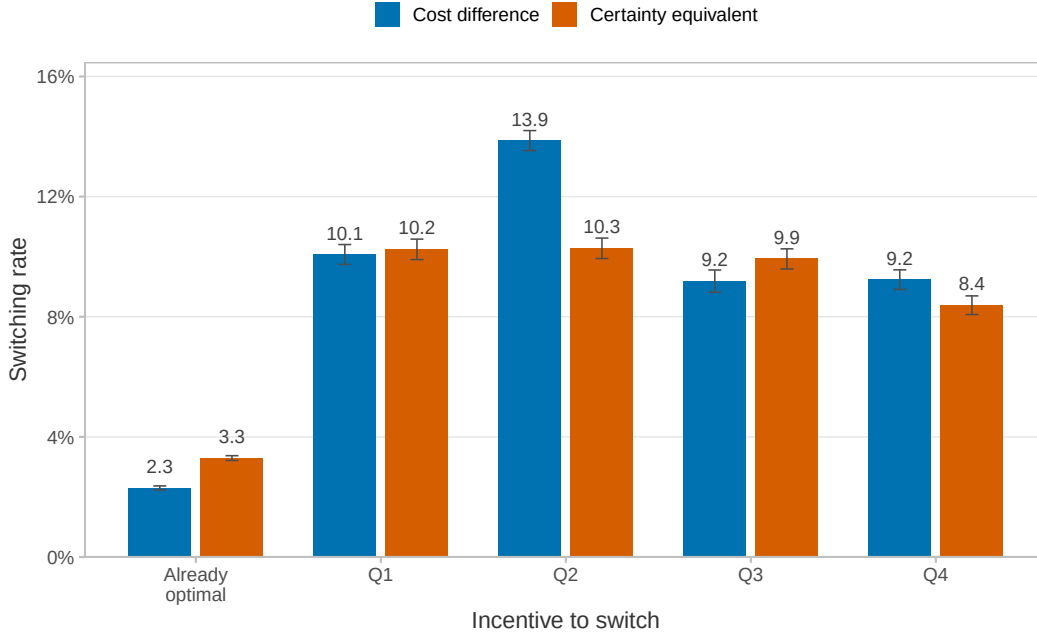
Taken together, these results indicate that inertia in deductible choice is partly—but not

Table 7: Switching and Incentives to Switch

	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Incentive measured as a cost difference</i>					
Incentive to switch (000 CHF)	0.068*** (0.002)	0.090*** (0.002)	0.090*** (0.002)	0.090*** (0.002)	0.082*** (0.003)
Chronic disease				0.001 (0.003)	
N	90 066	315 231	315 231	315 231	180 132
Within R^2	—	0.016	0.016	0.016	0.013
<i>Panel B. Incentive measured as a certainty equivalent</i>					
Incentive to switch (000 CHF)	0.072*** (0.002)	0.121*** (0.004)	0.120*** (0.004)	0.121*** (0.004)	0.076*** (0.005)
Chronic disease				0.006+ (0.003)	
N	90 066	315 231	315 231	315 231	180 132
Within R^2	—	0.010	0.009	0.010	0.004
Sample	Cross-section	Baseline panel	Baseline panel	Baseline panel	Shorter-tenured
Individual FE		✓	✓	✓	✓
Year FE			✓		

Notes: The dependent variable is an indicator for choosing a different deductible than in the previous year. The incentive to switch is measured in thousands of CHF. The *baseline panel* used in columns (2)–(4) retains those person-years for which the lagged deductible reflects a full year of enrollment: longer-tenured enrollees enter the panel in 2019 and shorter-tenured enrollees in 2020, since the 2018 cohort’s initial choices are made partway through their year of market entry (Section 3). Column (1) is a single-year cross-section for 2020—the first year in which both groups are in the baseline panel—and therefore has one observation per enrollee; column (5) restricts to the 2018 cohort and uses all of its person-years for which a lagged deductible is observed (2019–2022). Panel A measures the incentive as a difference in realized cost and Panel B as a difference in certainty equivalents, both as defined in the text. Columns (2)–(5) include individual fixed effects, which absorb the constant term. Standard errors are clustered at the individual level in columns (2)–(5) and are conventional in column (1).

Figure 5: Switching Rate and Incentive to Switch



Notes: This figure plots average switching rates by bin of the incentive to switch, for the baseline panel of Table 7 (315,231 enrollee-years). The leftmost pair shows enrollees whose default plan is already optimal (zero incentive); the remaining pairs correspond to quartiles of the strictly positive incentive. Blue bars measure the incentive as a difference in total cost to the enrollee (premium plus cost-sharing) between the enrollee’s default deductible and the cost-minimizing deductible. Orange bars measure it as a difference in certainty equivalents, as defined in the text. Bins are constructed separately for each measure, so their compositions differ. Whiskers are 95% confidence intervals. Switching does not rise with the incentive beyond the median under either measure, and under the risk-adjusted measure it is flat throughout, indicating that large financial stakes alone are insufficient to overcome inertia.

fully—explained by rational frictions. Even in settings with simple menus and transparent incentives, many consumers fail to adjust away from suboptimal defaults, resulting in persistent and economically meaningful losses.

5 Counterfactual Analysis and Policy Implications

The preceding analysis documents that consumer inertia in deductible choice is large, persistent, and driven by market tenure rather than cognitive decline. A natural question follows: what would happen if we could eliminate this inertia? The answer, as we show in this section, involves a profound micro-macro tradeoff. At the individual level, the choice architecture is failing consumers, costing them over CHF 2 billion annually—and financial incentives alone cannot wake them up. Only a smart default will. At the systemic level, however, those same inertial consumers are currently overpaying into the risk pool, and

fixing their individual mistakes would drain over a billion francs in implicit cross-subsidies, forcing premium increases for everyone. We quantify both sides of this dilemma through two counterfactual exercises. Both counterfactual calculations use all longer- and shorter-tenured enrollees aged 26 and above over 2019–2022. The details of the calculation is in Appendix B.

5.1 The Cost of Inertia: A Smart Default

We first aggregate foregone savings from suboptimal deductible choice. For each longer-tenured enrollee in year t , we calculate foregone savings as the difference in total cost to the enrollee (premium plus cost-sharing) between the enrollee’s chosen deductible and the cost-minimizing deductible under perfect foresight. This provides an upper bound on the gains from a “smart default” that automatically assigns each enrollee to their individually optimal plan, since an implementable rule would have to predict spending rather than observe it. The comparison is in realized cost alone and does not price the change in risk exposure that a different deductible entails.

The results are striking in their magnitude. In 2019, total foregone savings among longer-tenured enrollees amount to approximately CHF 2.5 billion, with an average loss of CHF 361 per enrollee per year. Including shorter-tenured enrollees raises the total to CHF 2.6 billion. While the aggregate loss declines modestly over time—to CHF 2.1 billion by 2022—it remains economically substantial throughout the observation period. On average, the Swiss market leaves roughly CHF 2.3 billion per year on the table due to inertial deductible choice, representing approximately 6–7% of total enrollee cost.

These numbers bound the money an intervention could recover: an algorithmic smart default that moved every enrollee to their cost-minimizing plan would recapture the bulk of this loss. Even under a more conservative counterfactual—assigning longer-tenured enrollees the same average choice quality as shorter-tenured enrollees—aggregate savings would amount to CHF 40–230 million annually.⁹

5.2 Risk Pool Equilibrium Effects

However, eliminating inertia does not come without systemic consequences. As [Handel \(2013\)](#) demonstrates, consumer inertia can stabilize the risk pool by preventing the adverse selection

⁹We calculate the difference between the average annual foregone savings for matched longer- and shorter-tenured enrollees, then multiply this number by the total number of longer-tenured enrollees in that year to obtain the estimates. The range reflects the fact that the difference in the average foregone savings between shorter- and longer-tenured enrollees narrows over time as shorter-tenured enrollees’ own choice quality deteriorates with market tenure.

that would arise if all consumers continuously optimized. When inertial consumers remain in plans with lower deductibles than they need, they pay higher premiums for coverage they do not use, effectively cross-subsidizing the system. Removing this inertia—however well-intentioned—risks unraveling these implicit transfers.

We investigate this mechanism directly in the Swiss market. For each longer-tenured enrollee, we compute the insurer’s margin under two scenarios: (i) the status quo, where the enrollee remains in their current plan, and (ii) the counterfactual, where the enrollee moves to their cost-minimizing plan. The insurer’s margin is defined as premium revenue minus payout (gross medical spending less the enrollee’s cost-sharing contribution).

The results reveal a substantial systemic deficit. In 2019, total insurer margins under the status quo are CHF 3.0 billion. Under the counterfactual where all enrollees optimize, margins fall by approximately CHF 1.1 billion. This deficit arises because over-insured enrollees (those on unnecessarily low deductibles) currently pay high premiums for coverage they do not use, effectively cross-subsidizing the system. We estimate that over-insured enrollees overpay by approximately CHF 1.6 billion annually relative to their individually optimal plans.¹⁰

If insurers were to maintain the same aggregate margin in a frictionless world, base premiums would need to increase by approximately CHF 130–165 per enrollee, or roughly 3–4%. This finding confirms, in a large-scale mandatory insurance market, the mechanism identified by [Handel \(2013\)](#): the same inertia that harms individual consumers also depresses premiums for the broader market. Policymakers considering smart defaults or mandatory re-enrollment must account for the equilibrium premium adjustment that would follow.

Our findings are consistent with recent empirical evidence from the Dutch health insurance market—which shares key structural features with Switzerland, including mandatory enrollment, annual deductible choice, and community-rated premiums. [Croes et al. \(2025\)](#) document that consumer inertia in the Dutch system counteracts adverse selection in deductible choice, providing direct empirical support for the cross-subsidization mechanism we quantify here. Together, the Swiss and Dutch evidence suggest that this tradeoff is a general feature of mandatory insurance markets with deductible choice, rather than an artifact of a particular institutional setting.

¹⁰The insurer margin under the status quo is $M_{it}^{sq} = P_{it} - (G_{it} - OOP_{it})$, where P_{it} is the enrollee’s annual premium, G_{it} is gross medical spending, and OOP_{it} is cost-sharing. The counterfactual margin is $M_{it}^{cf} = P_{it}^* - (G_{it} - OOP_{it}^*)$, where asterisks denote values at the cost-minimizing deductible. Premiums are recovered using the regulated rebate: the CHF 2,500 deductible carries an annual reduction of $(2,500 - 300) \times 0.70 = \text{CHF } 1,540$ relative to the CHF 300 baseline (KVV Art. 95 Abs. 2). The per-capita premium increase is $\Delta = (\sum M_{it}^{sq} - \sum M_{it}^{cf})/N_t$, and the percentage increase is $\Delta/\bar{P}_t$.

Taken together, the two counterfactual exercises reveal a fundamental policy dilemma. The micro-level problem is clear: the current choice architecture is failing individuals, and financial incentives—even when large—cannot reliably induce re-optimization. A smart default is the only intervention that would recapture the CHF 2.3 billion in annual foregone savings. However, the macro-level reality complicates any straightforward implementation. Because inertial consumers are currently overpaying through excessive premiums on dominated plans, they implicitly subsidize the entire risk pool by CHF 860 million–1.1 billion per year. Eliminating their mistakes—however well-intentioned—would drain these cross-subsidies and necessitate a base premium increase of approximately 3–4% to maintain market equilibrium. Any redesign of enrollment architecture must explicitly anticipate and manage this equilibrium adjustment, balancing gains to individual enrollees against the aggregate cost of unwinding the implicit subsidies that inertia currently provides. Because Switzerland’s basic insurance market features standardized benefits and transparent pricing, the average foregone savings of CHF 361 per enrollee in 2019 may serve as a useful benchmark, though direct comparisons to other markets should account for differences in product complexity, choice architecture, and supply-side institutions.

6 Conclusion

This paper provides evidence on how consumers choose deductibles in a regulated health insurance market where coverage is standardized and optimal cost-sharing is relatively easy to characterize. Using nationwide administrative data from Switzerland, we document persistent patterns of suboptimal deductible choice that arise not primarily from lack of access to information, but from limited re-optimization once an initial choice is made. We further show that the age gradient in inertia—widely attributed to cognitive decline—is driven almost entirely by market tenure, and we quantify the aggregate cost of inertia through counterfactual exercises. Our analysis is subject to data limitations: we do not observe enrollment histories prior to 2017 and therefore cannot determine whether the lower choice quality among longer-tenured enrollees reflects deterioration of initially appropriate choices or persistence of initially poor ones. Our comparison of shorter- and longer-tenured enrollees may also be affected by compositional differences, as shorter-tenured enrollees are predominantly recent immigrants. We address these concerns through matching on observable characteristics and by focusing on within-group patterns that are less sensitive to level differences between the two populations.

Our findings have several implications for insurance market design and, more broadly, for

the design of any market relying on recurring consumer choice under auto-renewal. First, the absence of learning in the short term among shorter-tenured enrollees suggests that policies relying on repeated choice opportunities or passive experience are unlikely to substantially improve consumer outcomes. Even when consumers initially make an active choice, they rarely revisit or revise that choice in subsequent years, despite accumulating information about their health risks. Second, the large and persistent foregone savings observed among longer-tenured enrollees highlight the importance of default rules. Because defaults anchor consumers progressively with market tenure—regardless of age or cognitive ability—inertia can undermine the risk-protection role of insurance precisely when it matters most.

Third, while switching responds to financial incentives, the response is weak and highly non-linear. Even large potential savings do not reliably trigger re-optimization. This limits the effectiveness of policies that rely solely on price signals or information disclosure to induce active choice. Instead, our results point to the potential value of interventions that directly address inertia, such as structured re-enrollment, default re-optimization, or mandatory active choice intervals triggered by market tenure rather than age.

Fourth, our counterfactual analysis reveals a fundamental policy tension. While a smart default would save consumers approximately CHF 2.3 billion annually, the implicit cross-subsidies generated by inertia currently stabilize insurer revenues, and their removal would necessitate a base premium increase of roughly 3–4%. Policymakers and platform designers must weigh the gains to individual enrollees from eliminating inertia against the systemic equilibrium consequences—a trade-off that extends well beyond health insurance to any subscription market where passive auto-renewal governs consumer behavior.

More broadly, our finding that inertia is a function of market tenure rather than biological age suggests that the “sinking effect” we document may extend to other recurring choice architectures, though the magnitude will depend on institutional specifics such as product complexity, switching costs, and supply-side frictions. The Swiss market, with its standardized benefits and transparent pricing, provides a setting where many common barriers to switching are absent, suggesting that the average foregone savings of CHF 361 per enrollee in 2019 may represent a lower bound relative to more complex choice environments. However, we caution against direct extrapolation, as other markets feature different information structures, stakes, and competitive dynamics. Understanding how contract design interacts with consumer behavior remains central to ensuring that markets relying on recurring choice deliver effective outcomes for consumers.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude (Anthropic) for language editing and for generating code used in parts of the data analysis. The authors reviewed and edited the output as needed and take full responsibility for the content of the published article.

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A Appendix: Formalizing the Sinking Effect

Throughout the main text, we document that consumer inertia intensifies with market tenure and refer to a “sinking effect” whereby enrollees become progressively anchored to their default plan. In this appendix, we formalize this phenomenon using survival analysis. Rather than simply reporting that switching rates are low, we model the precise dynamics of how—and how quickly—new market entrants lose their propensity to actively re-optimize.

A.1 Setup and Motivation

Our 2018 cohort provides a natural setting for survival analysis. All individuals entered the Swiss health insurance market simultaneously, so they share a common tenure clock that starts at zero. We observe them annually from 2019 through 2022, yielding up to four years of follow-up. The “event” of interest is the first deductible switch—the first time an enrollee actively changes their cost-sharing plan after initial enrollment. Enrollees who never switch during the observation window are right-censored at their last observed year.

This design addresses a key identification challenge: in cross-sectional data, it is impossible to distinguish whether low switching rates among long-tenured enrollees reflect a causal tenure effect or selection (i.e., individuals who are inherently passive self-selecting into long tenure). By tracking a single cohort from market entry, we observe the within-cohort evolution of switching behavior as tenure accumulates, holding the composition of the cohort fixed.

We proceed in two steps. First, we estimate Kaplan-Meier survival curves to visualize the probability that a shorter-tenured enrollee has not yet switched by each tenure year. Second, we compute discrete-time hazard rates to measure the annual conditional probability of first switching.

A.2 Kaplan-Meier Survival Curves

We define the survival function $S(t) = \Pr(\text{no switch by year } t)$ and estimate it nonparametrically using the Kaplan-Meier estimator. Figure 6 presents the results.

Panel A shows the overall survival curve. At market entry (2018), the survival probability is 1 by construction. By the end of the first year of tenure (2019), approximately 97% of shorter-tenured enrollees have not yet switched—only 2.7% make an active change in their first opportunity. By year 4 (2022), the survival probability remains above 85%, and the median survival time is never reached: more than half of all shorter-tenured enrollees *never*

switch their deductible over the four-year observation window. The curve’s shape is revealing. It declines gradually and approximately linearly, with no sharp drop-off at any particular tenure year. This implies that the transition into inertia is not a sudden event triggered by a specific experience; rather, consumers gradually and steadily lose engagement with their plan choice as tenure accumulates.

Panel B stratifies the survival curves by age group at entry. The curves exhibit broadly similar shapes across age groups, with all cohorts experiencing the same gradual sinking into their default plan. The 66+ group exhibits modestly higher survival (lower switching), while younger and middle-aged cohorts are closely clustered. The parallel decay across groups suggests that the sinking effect operates similarly regardless of age at entry.

A.3 Discrete-Time Hazard Rates

Because our data are observed at annual intervals, we compute discrete-time hazard rates: the probability of switching for the first time in year t , conditional on not having switched in any prior year. Formally, the hazard rate at tenure t is:

$$h(t) = \Pr(\text{first switch in year } t \mid \text{no switch before year } t) = \frac{d(t)}{n(t)}, \quad (3)$$

where $d(t)$ is the number of first-time switchers in year t and $n(t)$ is the number of enrollees still at risk (not yet switched and still observed).

Table 8 reports the results. The overall hazard rate starts at 2.7% in year 1 (2019), rises to 4.3% in year 2, and then plateaus at approximately 4.3–4.6% through year 4. Two features of this pattern are noteworthy. First, the hazard rates are uniformly low: even at their peak, fewer than 5% of at-risk enrollees switch in any given year. The median survival time is never reached, meaning that the typical shorter-tenured enrollee remains in their initial plan indefinitely. Second, the hazard stabilizes quickly. After a modest increase from year 1 to year 2—likely reflecting a learning period in which some enrollees discover that their initial choice was suboptimal—the switching rate plateaus at a level far below what would be expected if consumers were continuously optimizing.

To benchmark these rates, [Ketcham et al. \(2015\)](#) report annual switching of roughly 10–11% in Medicare Part D, a setting with substantially more supply-side friction: formulary changes, plan exits, and tier restructuring. Our annual hazard peaks below 5%. Swiss enrollees therefore re-optimize less than half as often as Part D enrollees, despite facing a fixed benefit package, no plan exits, and a menu of six options. The barrier is evidently not

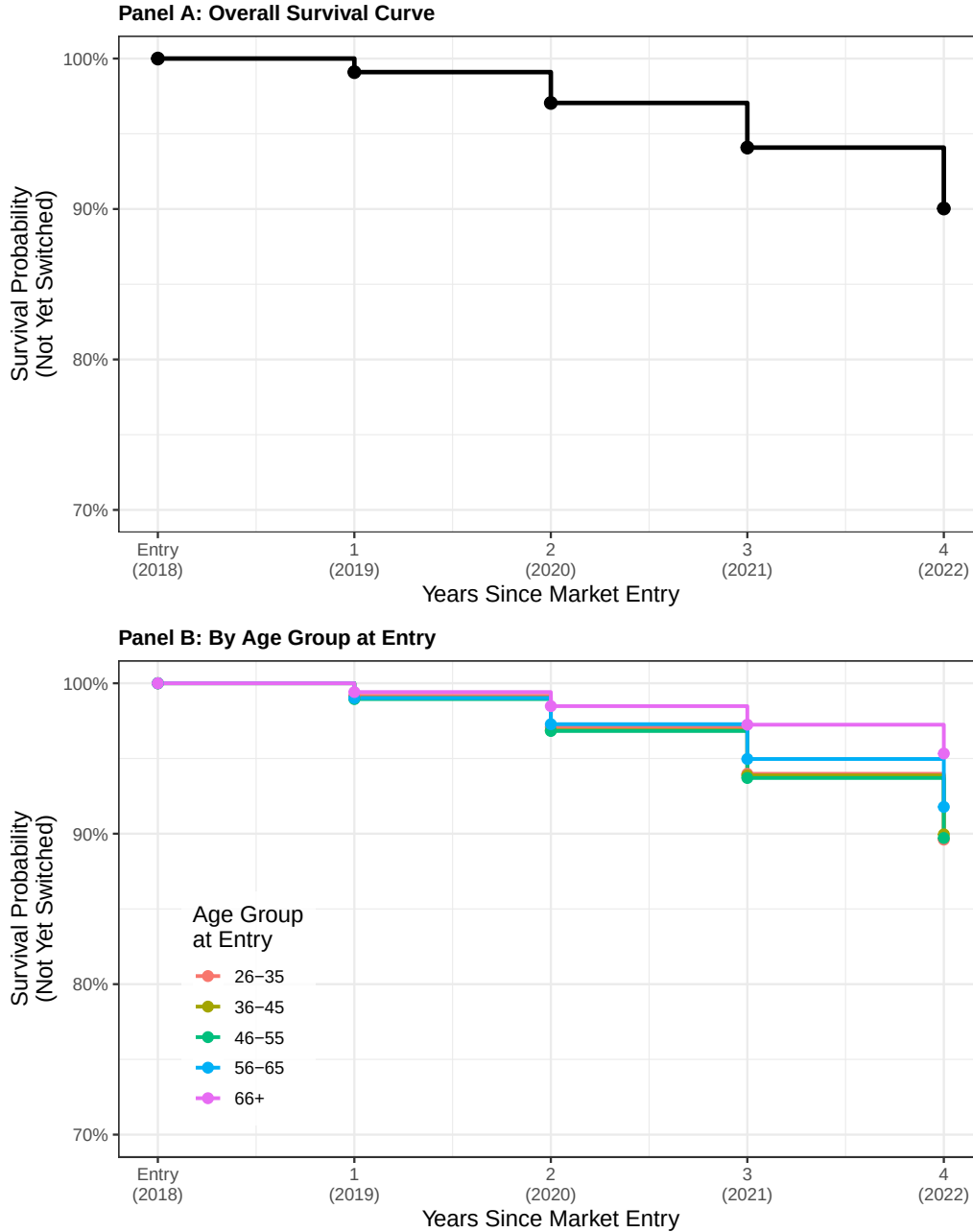


Figure 6: Kaplan-Meier Survival Curves: Time to First Deductible Switch

Notes: This figure plots Kaplan-Meier estimates of the survival function (probability of not having switched deductibles) for the 2018 cohort, tracked from 2019 through 2022. Panel A shows the overall survival curve. Panel B stratifies by age group at market entry. The event is defined as the first deductible switch. Enrollees who never switch are right-censored at their last observed year. $N = 639,890$ shorter-tenured enrollees; 57,818 events (9.0%).

the complexity of the alternatives. This suggests that the hazard rates we document reflect a behavioral baseline—the rate at which consumers re-optimize when the only barrier is their

Table 8: Discrete-Time Hazard Rates: Probability of First Switch by Tenure Year

Age Group	Year 1 (2019)	Year 2 (2020)	Year 3 (2021)	Year 4 (2022)
26-35	0.025	0.042	0.045	0.045
36-45	0.027	0.045	0.048	0.043
46-55	0.030	0.045	0.049	0.045
56-65	0.031	0.042	0.046	0.039
66+	0.021	0.025	0.026	0.023
All	0.027	0.043	0.046	0.043

Notes: This table reports discrete-time hazard rates for the 2018 cohort. Each cell shows the probability of switching deductibles for the first time in a given year, conditional on not having switched in any prior year. The at-risk population shrinks each year as switchers exit the risk set.

own inattention.

A.4 Discussion

The survival analysis formalizes two key properties of the sinking effect documented in the main text:

1. **Rapid onset.** About 85% of shorter-tenured enrollees never switch their deductible within four years of market entry. The transition from active chooser to passive default-holder occurs within the first one to two years of tenure, after which switching propensities stabilize at uniformly low levels. The cumulative switching rate of approximately 15% over four years is comparable in magnitude to the annual rates observed in Medicare Part D (Ketcham et al., 2015), despite the Swiss market’s far simpler and more transparent choice architecture. This suggests that the sinking effect we document reflects a behavioral baseline—the irreducible friction cost of recurring choice under auto-renewal defaults.
2. **Resilience to incentives.** The hazard rate plateaus at under 5% per year regardless of tenure, implying that the vast majority of enrollees remain anchored to their initial plan indefinitely. This pattern is consistent with the mechanism analysis in the main text: while financial incentives can induce marginal switching, the choice architecture itself—specifically, the auto-renewal default—is the binding constraint on consumer engagement.

These findings provide a behavioral micro-foundation for the inertia documented in the main analysis and reinforce the policy conclusion that interventions must directly address

the default mechanism—through mandatory active choice intervals, algorithmic default re-optimization, or structured re-enrollment—rather than relying on financial incentives or information provision to counteract the sinking effect.

B Appendix: Counterfactual Methodology

This appendix describes the construction of the counterfactual exercises reported in Section 5, so that a reader with access to the underlying data can reproduce all reported figures.

B.1 Data

The analysis uses individual-level administrative records from the Swiss Federal Office of Public Health (BAG), covering the full population enrolled in mandatory health insurance from 2017 to 2022. For each enrollee-year, we observe the approved monthly premium, the chosen deductible, annual gross medical spending, and total premiums paid. The sample is restricted to adult enrollees on the standard deductible menu (CHF 300 to CHF 2,500) over the period 2019–2022, yielding approximately 6.8 million longer-tenured enrollees per year.

B.2 Recovering Premiums at Every Deductible Level

We observe each enrollee’s monthly premium p_i at their chosen deductible d_i . The annual premium at the CHF 300 baseline is:

$$P_{300,i} = 12 \cdot p_i + (d_i - 300) \times 0.70. \quad (4)$$

This inverts the regulated rebate: an enrollee on a higher deductible pays a lower premium, and the difference is $(d_i - 300) \times 0.70$ per year. The premium at any deductible d is then:

$$P_{d,i} = P_{300,i} - (d - 300) \times 0.70. \quad (5)$$

B.3 Computing Out-of-Pocket Cost-Sharing

The out-of-pocket cost at deductible d given gross medical spending G_i is:

$$OOP_{d,i} = \begin{cases} G_i & \text{if } G_i \leq d, \\ d + 0.10 \times (G_i - d) & \text{if } d < G_i \leq d + 7,000, \\ d + 700 & \text{if } G_i > d + 7,000. \end{cases} \quad (6)$$

B.4 Identifying the Cost-Minimizing Deductible

The total cost to the enrollee under deductible d is $C_{d,i} = P_{d,i} + OOP_{d,i}$. The cost-minimizing deductible is:

$$d_i^* = \begin{cases} 300 & \text{if } C_{300,i} \leq C_{2500,i}, \\ 2,500 & \text{otherwise.} \end{cases} \quad (7)$$

B.5 Counterfactual Exercise 1: The Cost of Inertia

Foregone savings for enrollee i in year t is:

$$FS_{it} = C_{d_i,i} - C_{d_i^*,i} \geq 0. \quad (8)$$

In 2019, the average foregone savings among longer-tenured enrollees is CHF 361, with approximately 6.8 million adult enrollees in the sample. Aggregating yields the total foregone savings:

$$\text{Total foregone savings}_t = \sum_{i \in \text{existing}} FS_{it}. \quad (9)$$

For 2019, this sum amounts to CHF 2.5 billion. Including shorter-tenured enrollees raises the total to CHF 2.6 billion. Over the 2019–2022 period, the annual average is approximately CHF 2.3 billion, representing 6–7% of total enrollee spending. This exercise uses perfect foresight: we evaluate choice quality using realized spending in year t , providing an upper bound on the gains from a smart default.

B.6 Counterfactual Exercise 2: Risk Pool Equilibrium

For each longer-tenured enrollee, the insurer's margin under the status quo is:

$$M_{it}^{sq} = P_{it} - \underbrace{(G_{it} - OOP_{d_i,it})}_{\text{insurer payout}}, \quad (10)$$

where P_{it} is the observed premium, G_{it} is gross medical spending, and $OOP_{d_i,it}$ is cost-sharing at the chosen deductible. In 2019, the aggregate status quo margin across all longer-tenured enrollees is CHF 3.0 billion. This is a gross underwriting margin, not profit: it omits administrative costs and transfers under the federal risk-equalization scheme. The exercise uses only the change in the margin between the status quo and the counterfactual, and both omitted components are unchanged between the two. Risk equalization is computed from age, sex, prior-year hospitalization and pharmaceutical cost groups, none of which a deductible

switch alters, so transfers are identical across the scenarios rather than merely netting out in aggregate.

Under the counterfactual where the enrollee moves to the cost-minimizing deductible d_i^* :

$$M_{it}^{cf} = P_{d_i^*, it} - (G_{it} - OOP_{d_i^*, it}). \quad (11)$$

When all enrollees optimize, many shift to the CHF 2,500 deductible, lowering their premiums and increasing their cost-sharing. This reduces insurer premium revenue while also reducing insurer payouts. On net, the counterfactual aggregate margin in 2019 falls to CHF 1.8 billion. Gross medical spending G_{it} is held constant—we do not model moral hazard responses to the change in cost-sharing.

The systemic deficit is the aggregate margin loss from universal optimization:

$$\Delta_t = \sum_i M_{it}^{cf} - \sum_i M_{it}^{sq} < 0. \quad (12)$$

In 2019, $\Delta_t \approx -\text{CHF } 1.1 \text{ billion}$. The deficit declines modestly over time to CHF 860 million by 2022 as the composition of enrollee spending shifts. To maintain the same aggregate margin, the required per-capita premium increase is:

$$\delta_t = \frac{|\Delta_t|}{N_t}, \quad \text{percentage increase} = \frac{\delta_t}{\bar{P}_t} \times 100, \quad (13)$$

where $\bar{P}_t$ is the average annual premium. With approximately 6.8 million longer-tenured enrollees and an average annual premium of CHF 4,182 in 2019, this yields a per-capita increase of CHF 163, or 3.9%. Across the 2019–2022 period, the required increase ranges from CHF 129 to CHF 165 per enrollee (3.1–3.9%).

B.7 Interpretation and Limitations

Three features of this methodology merit discussion. First, the perfect-foresight assumption in Exercise 1 provides an upper bound on foregone savings; in practice, a smart default would need to predict future spending, which would yield somewhat smaller gains. Second, Exercise 2 holds gross spending constant, ignoring potential moral hazard responses to deductible changes. If enrollees who move to higher deductibles reduce their utilization, insurer payouts would fall further, partially offsetting the systemic deficit. Third, both

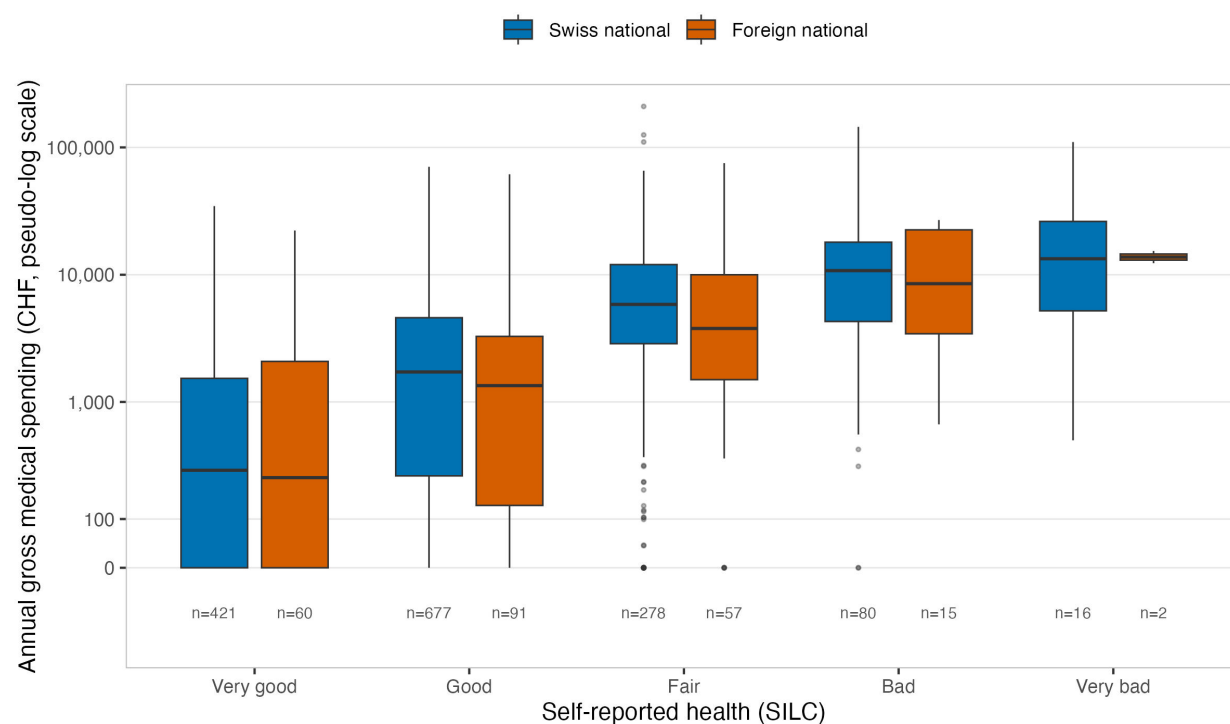
exercises treat the 70% rebate rate as a binding constraint. In practice, some insurers offer rebates below the regulatory maximum; our calculations therefore represent the case where premium differentials across deductibles are at their largest.

C Appendix: Medical Spending by Health Status and Nationality

This appendix uses a subsample of enrollees for whom the administrative records are linked to the Swiss Survey on Income and Living Conditions (SILC) to examine whether foreign and Swiss nationals with the same self-reported health have similar medical spending. The linked file covers survey waves 2017 through 2020 and reports, on the same record, the SILC general-health item (very good, good, fair, bad, very bad), nationality, and the enrollee’s annual gross medical spending from the insurer. Restricting to adults with non-missing health and nationality leaves 1,697 enrollee-years, of which 1,472 are Swiss nationals and 225 are foreign nationals.

Figure 7 plots the distribution of annual gross medical spending for each health category, separately by nationality. Spending rises steeply with worse self-reported health in both groups: the median moves from below CHF 300 among those in very good health to above CHF 10,000 among those in bad or very bad health, and the share with zero spending falls from more than a third to essentially zero. Within each health category, the two nationality groups have similar distributions. Foreign nationals have a somewhat lower median in the good, fair, and bad categories, but the interquartile ranges overlap substantially, and the differences are small relative to the gradient across health categories. The very bad category contains only two foreign nationals and is shown for completeness only.

Figure 7: Medical Spending by Self-Reported Health and Nationality, SILC-Linked Sample



Notes: SILC-linked administrative subsample, survey waves 2017–2020, adults with non-missing self-reported health and nationality. $N = 1,697$ enrollee-years (1,472 Swiss nationals, 225 foreign nationals); cell sizes are printed below each box. Boxes show the interquartile range and median; whiskers extend to 1.5 times the interquartile range; points are outliers. The vertical axis is a pseudo-logarithmic scale (linear near zero, logarithmic above) so that enrollees with zero spending remain on the plot.